

# Node allocator benchmark: the whole matrix

10 builds (macro combinations) x 5 allocators x 5 pointer types x 8 scenarios, measured in one run

Measured 2026-09-29 21:32 to 00:02, Intel(R) Core(TM) i7-4700HQ CPU @ 2.40GHz (8 logical CPUs), Linux 5.15.0-194-generic, clang version 23.0.0git, glibc 2.35; -O2. Median of 5 runs; every measurement in a fresh process, after the CPU has cooled. Tables: ALLOCATOR\_BENCHMARK.md.

## Findings

- The default, `page_allocator_by_size`, is fastest or within 10% in 114 of 140 cells.
- `boost::pool_allocator` is quadratic on release: 85-144 us per object at 100,000 objects.
- Thread support is `root_ptr`'s largest cost: churn 58.0 -> 9.4 ns without it (6.2x).
- Striped locks: `root_ptr` with 4 threads 10.5x faster, single-threaded 1.5-2.6x slower.
- Thread caches: 4 threads 183 -> 3.4 ns per `unique_ptr`; churn 1.3x-3.1x faster.
- Zeroization (`BOOST_ZEROIZATION`) costs nothing measurable: 0.97-1.01x.
- `root_ptr` vs. `unique_ptr` / `shared_ptr`: churn 3.1x / 2.6x with threads, 1.6x / 1.0x without.
- `shared_node_ptr`: 1.06-1.15x `root_ptr` single-threaded, 1.6x-25x faster with 4 threads.
- `page_allocator_by_type` with 32 sparse types: 2.1x-4.3x the time and +1.6-1.9 MB.
- Noise: the 5 runs of a cell differ by 1.21x (median); under ~10% is not a difference.

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# Findings

*numbers are medians of 5 runs; ns per operation unless noted*

- The default allocator, `page_allocator_by_size`, is the fastest or within 10% of the fastest in 114 of the 140 measured (build, pointer, scenario) cells. It loses by a wide margin only with 4 threads and no thread caches, where every allocation takes the page pool's mutex: `std::allocator` (glibc, per-thread arenas) is 36x faster for `unique_ptr` (5.0 vs. 182.9 ns), 28x for `shared_ptr`, 19x for `shared_node_ptr`, and 1.24x for `root_ptr` behind its global lock (313 vs. 389 ns).
- `boost::pool_allocator` is quadratic when many objects are released: its `deallocate` keeps the free list sorted (`ordered_free`). Releasing 100,000 objects together costs 85-144 us per object for every pointer and build, and about 2x more each time the count doubles (189-402 us at 200,000). Every other allocator stays at 41-255 ns in every build. The allocator alone shows it too: a batch costs 70.7 us per block against 28.9 ns.
- Thread support (the default) is `root_ptr`'s largest single cost: with `BOOST_DISABLE_THREADS` and `page_allocator_by_size`, churn drops from 58.0 to 9.4 ns (6.2x), cycles from 139.6 to 50.9 ns per node (2.7x) and bulk release from 133.4 to 74.8 ns (1.8x): the global `root_ptr` mutex, atomic counts and the pool mutex go away. `shared_node_ptr` churn drops 2.2x (61.8 to 28.7 ns), `unique_ptr` 3.2x (18.6 to 5.8 ns, the page pool's mutex); `std::allocator` itself does not change (16.3 vs. 15.8 ns).
- Striped locks (`BOOST_ROOT_PTR_STRIPED_LOCKS`) trade single-threaded speed for scaling. With 4 threads, `root_ptr` goes from 313 to 29.7 ns per object with `std::allocator` (10.5x), and from 309 to 30.2 ns with `page_allocator_by_size` plus thread caches (10.2x); without the caches the page pool's mutex caps the gain at 1.36x. Single-threaded every `root_ptr` scenario is slower: 1.5x to 2.6x (geometric means), e.g. churn 58.0 to 120.2 ns and cycles 139.6 to 451.1 ns. `shared_node_ptr` takes no `root_ptr` lock and does not change (0.99-1.04x).
- Per-thread caches (`BOOST_PAGE_ALLOCATOR_THREAD_CACHE`) take the page pool's mutex off the common path. With 4 threads and `page_allocator_by_size`: `unique_ptr` 182.9 to 3.4 ns, `shared_ptr` 166.5 to 4.3 ns, `shared_node_ptr` 238.9 to 12.4 ns - now level with or faster than `std::allocator` (5.1, 5.8, 15.2 ns); `root_ptr` needs striped locks as well (above). Churn: the allocator alone 3.1x faster (17.1 to 5.5 ns), `unique_ptr` 2.9x, `shared_ptr` 2.7x, `shared_node_ptr` 1.6x, `root_ptr` 1.3x. Bulk, mixed, types, cycles and resident memory do not change (0.99-1.04x).
- Zeroization (`BOOST_ZEROIZATION`) costs nothing measurable: the geometric mean of time without / with it is 0.97-1.01x for every pointer and scenario, and single cells scatter within run-to-run noise (0.72-1.22x). Clearing a 72-248 byte node is small next to a 50-150 ns release. (Checked separately that the macro is active: a released node reads back as zeros.)
- `root_ptr` against the standard pointers (`page_allocator_by_size`): churn costs 3.1x `unique_ptr` and 2.6x `shared_ptr` with thread support (58.0 vs. 18.6 and 22.7 ns), 1.6x and 1.0x without (9.4 vs. 5.8 and 9.3 ns). It holds 100,000 objects in 12.5 MB against 7.0 MB (`unique_ptr`) and 9.3 MB (`shared_ptr`). It is the only one of them that reclaims cycles.
- `shared_node_ptr` (what `FCXXSS_SHARED_PTR` emits) against `root_ptr`, both on `page_allocator_by_size`: single-threaded with thread support it costs 1.06-1.15x as much (churn 61.8 vs. 58.0 ns) and holds the same objects in 19-25% more memory (a control block per object on top of the node). With 4 threads it has no global lock: 1.6x faster (239 vs. 389 ns), 24x with `std::allocator` and 25x with the thread caches (12.4 vs. 309 ns). Without thread support it is the slower one: churn 3.1x (28.7 vs. 9.4 ns). A cycle through it is never released.
- `page_allocator_by_type` costs more with many sparse types: 32 types of 100 objects each take 2.1x to 4.3x the time of `page_allocator_by_size` and 1.6-1.9 MB more memory (3.8-4.2 MB against 1.9-2.3 MB), because each type fills a 64 KiB page of its own.
- Noise: the slowest of a cell's 5 runs is 1.21x its fastest at the median (1.36x at the 90th percentile, up to 4x in the 4-thread scenario), so the medians are reported, and differences below about 10% are not significant. The run took 2.5 hours on a shared desktop machine.

## Setup, builds and how to read the matrices

| build                             | no threads | striped | caches | zero | pointers measured         |
|-----------------------------------|------------|---------|--------|------|---------------------------|
| threads (default)                 |            |         |        |      | all                       |
| threads + striped                 |            | yes     |        |      | root_ptr, shared_node_ptr |
| threads + caches                  |            |         | yes    |      | all                       |
| threads + striped + caches        |            | yes     | yes    |      | root_ptr, shared_node_ptr |
| threads + zero                    |            |         |        | yes  | root_ptr, shared_node_ptr |
| threads + striped + zero          |            | yes     |        | yes  | root_ptr, shared_node_ptr |
| threads + caches + zero           |            |         | yes    | yes  | root_ptr, shared_node_ptr |
| threads + striped + caches + zero |            | yes     | yes    | yes  | root_ptr, shared_node_ptr |
| no threads                        | yes        |         |        |      | all                       |
| no threads + zero                 | yes        |         |        | yes  | root_ptr, shared_node_ptr |

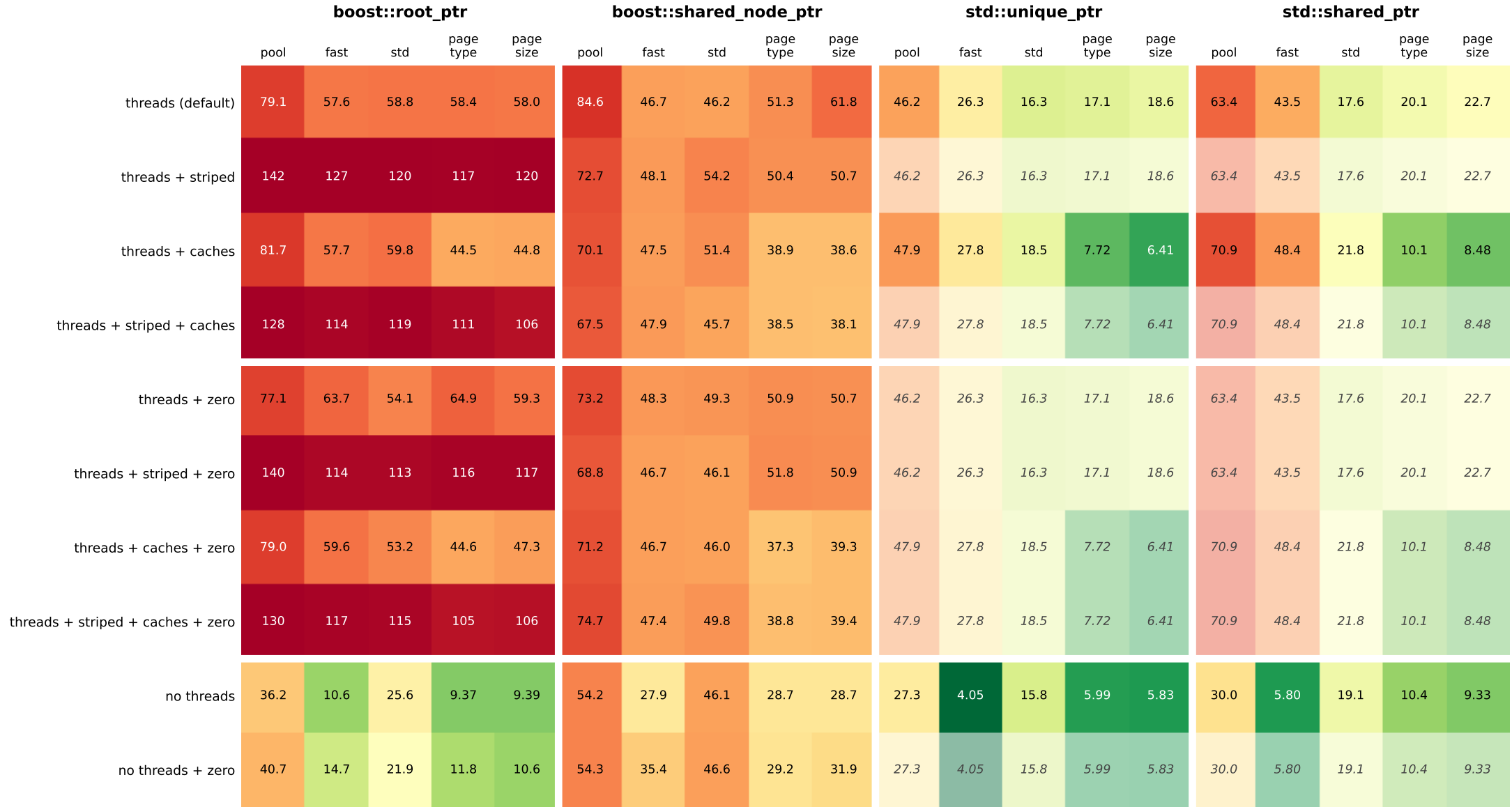
Macros. `BOOST_DISABLE_THREADS`: no `root_ptr` lock, plain reference counts, no pool mutexes.  
`BOOST_ROOT_PTR_STRIPED_LOCKS`: 1024 address-striped spin locks instead of `root_ptr`'s global mutex.  
`BOOST_PAGE_ALLOCATOR_THREAD_CACHE`: each thread keeps up to 16 KiB of free blocks of every page pool.  
`BOOST_ZEROIZATION`: every node is cleared (`memset`) before its memory is released.  
Striped locks and thread caches are inactive without threads, hence 8 + 2 builds. `BOOST_NO_EXCEPTIONS` only changes `root_ptr`'s static and dynamic cast constructors, which no scenario uses.

Pointers. `root_ptr`: `root_ptr<T>(x, new node<T, A<T>>(..));` `shared_node_ptr` (`FCXXSS_SHARED_PTR`): a `std::shared_ptr` owning the same node; `shared_ptr`: `std::allocate_shared`; `unique_ptr`: an `allocate_unique` with a deleter freeing through the same allocator; `allocator alone`: one node-sized block, no pointer. `unique_ptr`, `shared_ptr` and the `allocator alone` are measured in the builds where their code can differ (threads, threads + caches, no threads); faded italic cells carry those values to the equivalent builds.

Reading. Time and memory pages: rows are builds, columns allocators grouped by pointer; green = less.  
Effect pages: time without / time with one setting; green (above 1.00x) = the setting is faster.  
Every page says whether lower or higher is better. Colours are per page; a log scale throughout.

# Time per operation: churn (lower is better)

create and drop one 48-byte object, 1,000,000 times - ns per operation, median of 5 runs



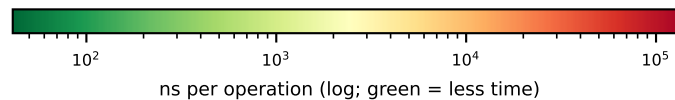
ns per operation (log; green = less time)

threads: built with thread support (the default) no threads: BOOST\_DISABLE\_THREADS striped: BOOST\_ROOT\_PTR\_STRIPED\_LOCKS caches: BOOST\_PAGE\_ALLOCATOR\_THREAD\_CACHE zero: BOOST\_ZEROIZATION  
Faded italic cells: unique\_ptr, shared\_ptr and the allocator alone have no node and no root\_ptr lock - striped locks and zeroization do not change their code - so they show the equivalent build's measurement.

# Time per operation: bulk (lower is better)

100,000 live 48-byte objects, released together - ns per operation, median of 5 runs

|                                   | boost::root_ptr |      |      |           |           | boost::shared_node_ptr |      |     |           |           | std::unique_ptr |      |      |           |           | std::shared_ptr |      |      |           |           |
|-----------------------------------|-----------------|------|------|-----------|-----------|------------------------|------|-----|-----------|-----------|-----------------|------|------|-----------|-----------|-----------------|------|------|-----------|-----------|
|                                   | pool            | fast | std  | page type | page size | pool                   | fast | std | page type | page size | pool            | fast | std  | page type | page size | pool            | fast | std  | page type | page size |
| threads (default)                 | 141,871         | 138  | 144  | 131       | 133       | 144,252                | 151  | 150 | 147       | 142       | 85,265          | 63.0 | 65.9 | 47.5      | 47.6      | 136,876         | 99.7 | 77.2 | 64.2      | 63.5      |
| threads + striped                 | 141,697         | 240  | 252  | 235       | 241       | 142,903                | 150  | 151 | 142       | 143       | 85,265          | 63.0 | 65.9 | 47.5      | 47.6      | 136,876         | 99.7 | 77.2 | 64.2      | 63.5      |
| threads + caches                  | 140,769         | 137  | 138  | 131       | 131       | 142,988                | 151  | 156 | 150       | 145       | 84,610          | 62.2 | 65.8 | 44.2      | 42.9      | 138,142         | 100  | 79.4 | 61.2      | 61.3      |
| threads + striped + caches        | 143,332         | 249  | 250  | 240       | 233       | 141,926                | 150  | 149 | 147       | 148       | 84,610          | 62.2 | 65.8 | 44.2      | 42.9      | 138,142         | 100  | 79.4 | 61.2      | 61.3      |
| threads + zero                    | 140,945         | 138  | 144  | 131       | 129       | 143,772                | 155  | 155 | 142       | 142       | 85,265          | 63.0 | 65.9 | 47.5      | 47.6      | 136,876         | 99.7 | 77.2 | 64.2      | 63.5      |
| threads + striped + zero          | 142,621         | 244  | 255  | 234       | 232       | 143,751                | 151  | 155 | 142       | 143       | 85,265          | 63.0 | 65.9 | 47.5      | 47.6      | 136,876         | 99.7 | 77.2 | 64.2      | 63.5      |
| threads + caches + zero           | 143,739         | 134  | 144  | 137       | 135       | 145,759                | 149  | 154 | 148       | 150       | 84,610          | 62.2 | 65.8 | 44.2      | 42.9      | 138,142         | 100  | 79.4 | 61.2      | 61.3      |
| threads + striped + caches + zero | 143,132         | 246  | 250  | 241       | 236       | 144,356                | 151  | 150 | 150       | 145       | 84,610          | 62.2 | 65.8 | 44.2      | 42.9      | 138,142         | 100  | 79.4 | 61.2      | 61.3      |
| no threads                        | 145,369         | 81.2 | 91.8 | 74.8      | 74.8      | 142,011                | 125  | 141 | 116       | 116       | 88,068          | 42.3 | 64.8 | 41.4      | 40.9      | 139,020         | 58.8 | 80.0 | 55.9      | 55.3      |
| no threads + zero                 | 145,919         | 84.8 | 89.5 | 76.9      | 76.4      | 144,348                | 123  | 137 | 119       | 114       | 88,068          | 42.3 | 64.8 | 41.4      | 40.9      | 139,020         | 58.8 | 80.0 | 55.9      | 55.3      |

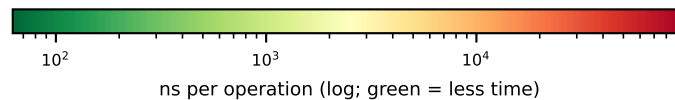


threads: built with thread support (the default) no threads: BOOST\_DISABLE\_THREADS striped: BOOST\_ROOT\_PTR\_STRIPED\_LOCKS caches: BOOST\_PAGE\_ALLOCATOR\_THREAD\_CACHE zero: BOOST\_ZEROIZATION  
Faded italic cells: unique\_ptr, shared\_ptr and the allocator alone have no node and no root\_ptr lock - striped locks and zeroization do not change their code - so they show the equivalent build's measurement.

# Time per operation: mixed (lower is better)

50,000 each of 16, 48 and 200 bytes, interleaved, released together - ns per operation, median of 5 runs

|                                   | boost::root_ptr |      |     |           |           | boost::shared_node_ptr |      |     |           |           | std::unique_ptr |      |      |           |           | std::shared_ptr |      |     |           |           |
|-----------------------------------|-----------------|------|-----|-----------|-----------|------------------------|------|-----|-----------|-----------|-----------------|------|------|-----------|-----------|-----------------|------|-----|-----------|-----------|
|                                   | pool            | fast | std | page type | page size | pool                   | fast | std | page type | page size | pool            | fast | std  | page type | page size | pool            | fast | std | page type | page size |
| threads (default)                 | 99,512          | 170  | 178 | 137       | 155       | 97,293                 | 186  | 193 | 155       | 176       | 83,971          | 91.5 | 93.8 | 63.0      | 72.6      | 96,200          | 127  | 122 | 89.7      | 91.2      |
| threads + striped                 | 97,170          | 267  | 277 | 243       | 258       | 98,320                 | 184  | 229 | 154       | 156       | 83,971          | 91.5 | 93.8 | 63.0      | 72.6      | 96,200          | 127  | 122 | 89.7      | 91.2      |
| threads + caches                  | 95,371          | 162  | 182 | 144       | 142       | 98,280                 | 180  | 209 | 163       | 155       | 81,330          | 84.2 | 91.3 | 70.7      | 70.0      | 95,502          | 121  | 104 | 90.2      | 90.5      |
| threads + striped + caches        | 96,602          | 252  | 281 | 254       | 242       | 97,900                 | 159  | 214 | 164       | 160       | 81,330          | 84.2 | 91.3 | 70.7      | 70.0      | 95,502          | 121  | 104 | 90.2      | 90.5      |
| threads + zero                    | 98,230          | 166  | 154 | 144       | 144       | 99,161                 | 180  | 199 | 173       | 165       | 83,971          | 91.5 | 93.8 | 63.0      | 72.6      | 96,200          | 127  | 122 | 89.7      | 91.2      |
| threads + striped + zero          | 103,182         | 256  | 272 | 239       | 245       | 97,605                 | 159  | 201 | 149       | 162       | 83,971          | 91.5 | 93.8 | 63.0      | 72.6      | 96,200          | 127  | 122 | 89.7      | 91.2      |
| threads + caches + zero           | 98,105          | 160  | 180 | 156       | 147       | 99,258                 | 175  | 209 | 166       | 170       | 81,330          | 84.2 | 91.3 | 70.7      | 70.0      | 95,502          | 121  | 104 | 90.2      | 90.5      |
| threads + striped + caches + zero | 95,402          | 256  | 279 | 250       | 265       | 97,778                 | 156  | 182 | 157       | 156       | 81,330          | 84.2 | 91.3 | 70.7      | 70.0      | 95,502          | 121  | 104 | 90.2      | 90.5      |
| no threads                        | 101,238         | 112  | 129 | 98.9      | 87.1      | 105,652                | 156  | 199 | 145       | 126       | 82,876          | 70.8 | 93.4 | 62.4      | 66.8      | 96,604          | 88.6 | 116 | 82.7      | 82.7      |
| no threads + zero                 | 100,616         | 117  | 139 | 103       | 104       | 99,523                 | 155  | 199 | 148       | 146       | 82,876          | 70.8 | 93.4 | 62.4      | 66.8      | 96,604          | 88.6 | 116 | 82.7      | 82.7      |



threads: built with thread support (the default) no threads: BOOST\_DISABLE\_THREADS striped: BOOST\_ROOT\_PTR\_STRIPED\_LOCKS caches: BOOST\_PAGE\_ALLOCATOR\_THREAD\_CACHE zero: BOOST\_ZEROIZATION  
Faded italic cells: unique\_ptr, shared\_ptr and the allocator alone have no node and no root\_ptr lock - striped locks and zeroization do not change their code - so they show the equivalent build's measurement.

# Time per operation: types (lower is better)

32 types of equal size, 100 live objects of each - ns per operation, median of 5 runs

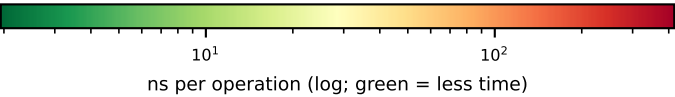
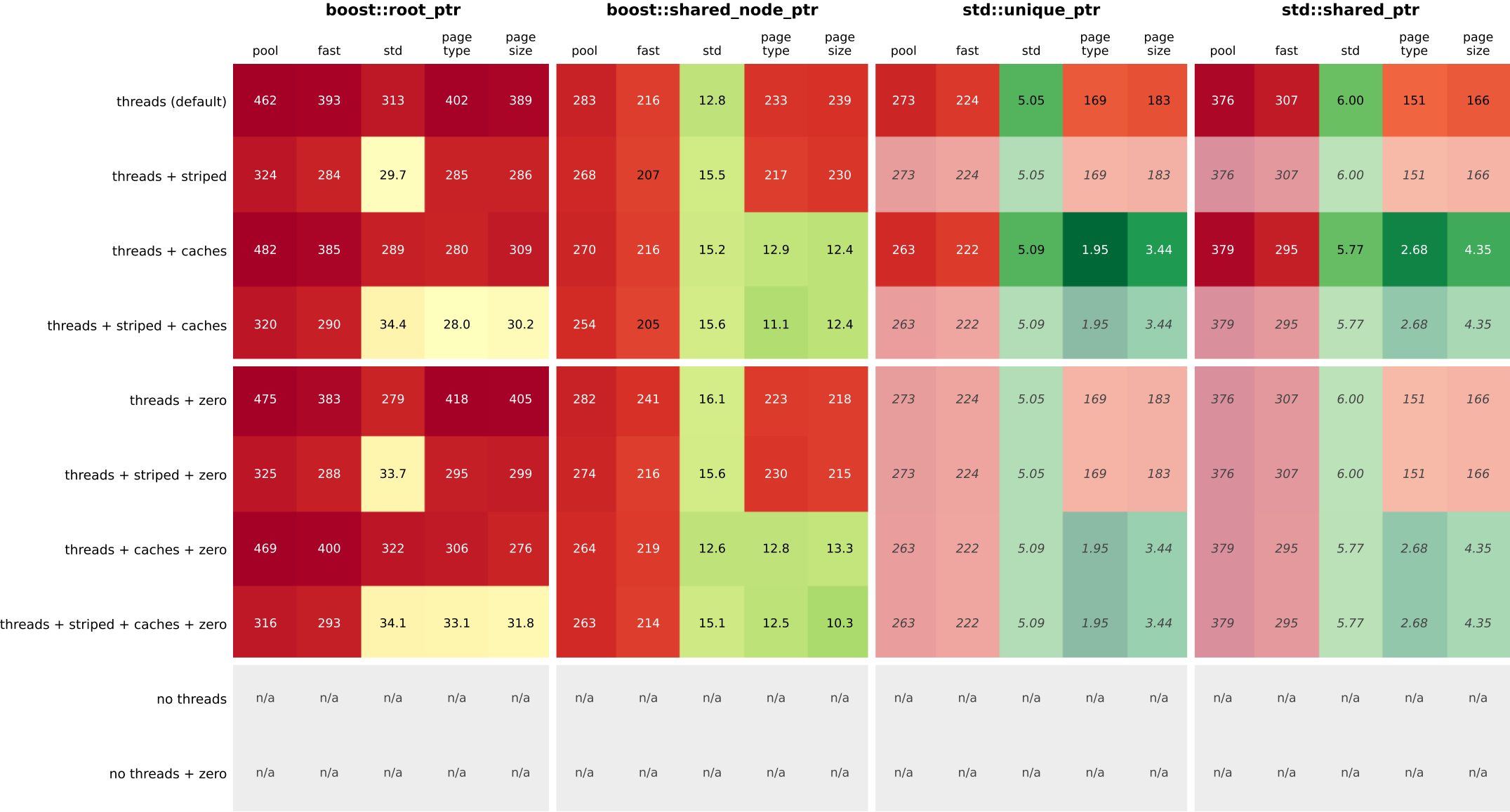
|                                   |  | boost::root_ptr |      |      |           |           | boost::shared_node_ptr |      |     |           |           | std::unique_ptr |      |      |           |           | std::shared_ptr |      |      |           |           |
|-----------------------------------|--|-----------------|------|------|-----------|-----------|------------------------|------|-----|-----------|-----------|-----------------|------|------|-----------|-----------|-----------------|------|------|-----------|-----------|
|                                   |  | pool            | fast | std  | page type | page size | pool                   | fast | std | page type | page size | pool            | fast | std  | page type | page size | pool            | fast | std  | page type | page size |
| threads (default)                 |  | 239             | 141  | 160  | 371       | 146       | 249                    | 168  | 169 | 346       | 168       | 170             | 74.5 | 75.1 | 283       | 66.0      | 204             | 112  | 88.2 | 276       | 85.8      |
| threads + striped                 |  | 339             | 281  | 273  | 443       | 269       | 263                    | 170  | 173 | 344       | 168       | 170             | 74.5 | 75.1 | 283       | 66.0      | 204             | 112  | 88.2 | 276       | 85.8      |
| threads + caches                  |  | 249             | 141  | 159  | 327       | 143       | 256                    | 148  | 176 | 355       | 168       | 177             | 73.4 | 76.6 | 287       | 63.2      | 208             | 97.6 | 89.8 | 310       | 76.2      |
| threads + striped + caches        |  | 343             | 251  | 278  | 504       | 262       | 252                    | 159  | 175 | 388       | 154       | 177             | 73.4 | 76.6 | 287       | 63.2      | 208             | 97.6 | 89.8 | 310       | 76.2      |
| threads + zero                    |  | 257             | 166  | 160  | 354       | 152       | 255                    | 171  | 176 | 352       | 164       | 170             | 74.5 | 75.1 | 283       | 66.0      | 204             | 112  | 88.2 | 276       | 85.8      |
| threads + striped + zero          |  | 350             | 284  | 273  | 437       | 265       | 251                    | 167  | 174 | 389       | 162       | 170             | 74.5 | 75.1 | 283       | 66.0      | 204             | 112  | 88.2 | 276       | 85.8      |
| threads + caches + zero           |  | 244             | 161  | 160  | 378       | 145       | 255                    | 173  | 169 | 385       | 165       | 177             | 73.4 | 76.6 | 287       | 63.2      | 208             | 97.6 | 89.8 | 310       | 76.2      |
| threads + striped + caches + zero |  | 338             | 288  | 281  | 503       | 272       | 267                    | 170  | 175 | 396       | 158       | 177             | 73.4 | 76.6 | 287       | 63.2      | 208             | 97.6 | 89.8 | 310       | 76.2      |
| no threads                        |  | 191             | 104  | 105  | 317       | 83.2      | 244                    | 150  | 156 | 363       | 138       | 162             | 55.3 | 72.9 | 297       | 57.7      | 172             | 68.8 | 88.9 | 306       | 74.7      |
| no threads + zero                 |  | 215             | 101  | 92.6 | 322       | 90.2      | 243                    | 151  | 142 | 366       | 138       | 162             | 55.3 | 72.9 | 297       | 57.7      | 172             | 68.8 | 88.9 | 306       | 74.7      |



threads: built with thread support (the default)   no threads: BOOST\_DISABLE\_THREADS   striped: BOOST\_ROOT\_PTR\_STRIPED\_LOCKS   caches: BOOST\_PAGE\_ALLOCATOR\_THREAD\_CACHE   zero: BOOST\_ZEROIZATION  
Faded italic cells: unique\_ptr, shared\_ptr and the allocator alone have no node and no root\_ptr lock - striped locks and zeroization do not change their code - so they show the equivalent build's measurement.

Time per operation: threads (lower is better)

4 threads, each creating and dropping 250,000 objects (wall clock / all operations) - ns per operation, median of 5 runs

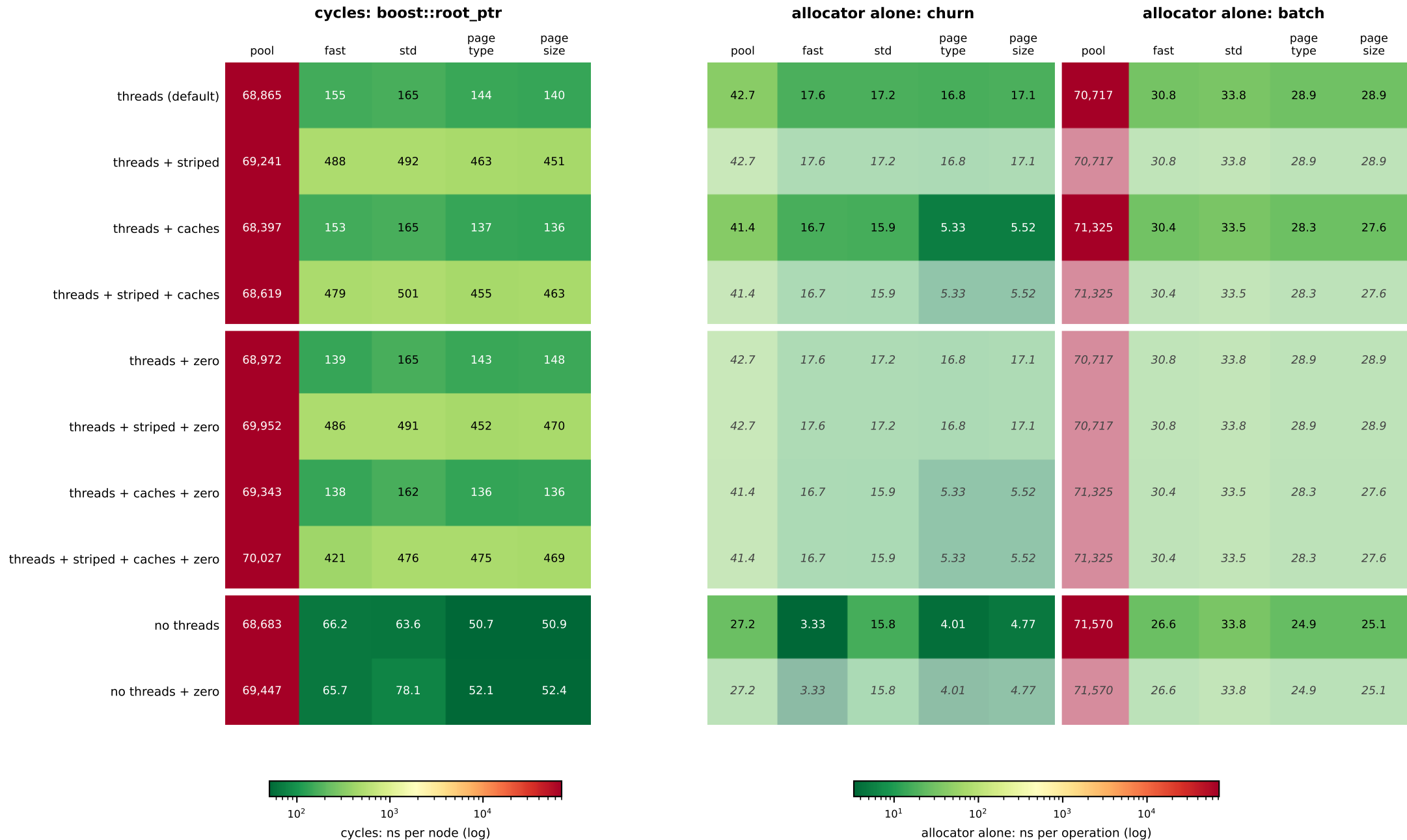


threads: built with thread support (the default) no threads: BOOST\_DISABLE\_THREADS striped: BOOST\_ROOT\_PTR\_STRIPED\_LOCKS caches: BOOST\_PAGE\_ALLOCATOR\_THREAD\_CACHE zero: BOOST\_ZEROIZATION  
Faded italic cells: unique\_ptr, shared\_ptr and the allocator alone have no node and no root\_ptr lock - striped locks and zeroization do not change their code - so they show the equivalent build's measurement.

## Time per operation: cycles, and the allocator alone (lower is better)

*cycles: 25,000 two-node cycles, reclaimed with their proxy (root\_ptr only), ns per node.*

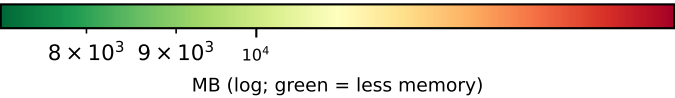
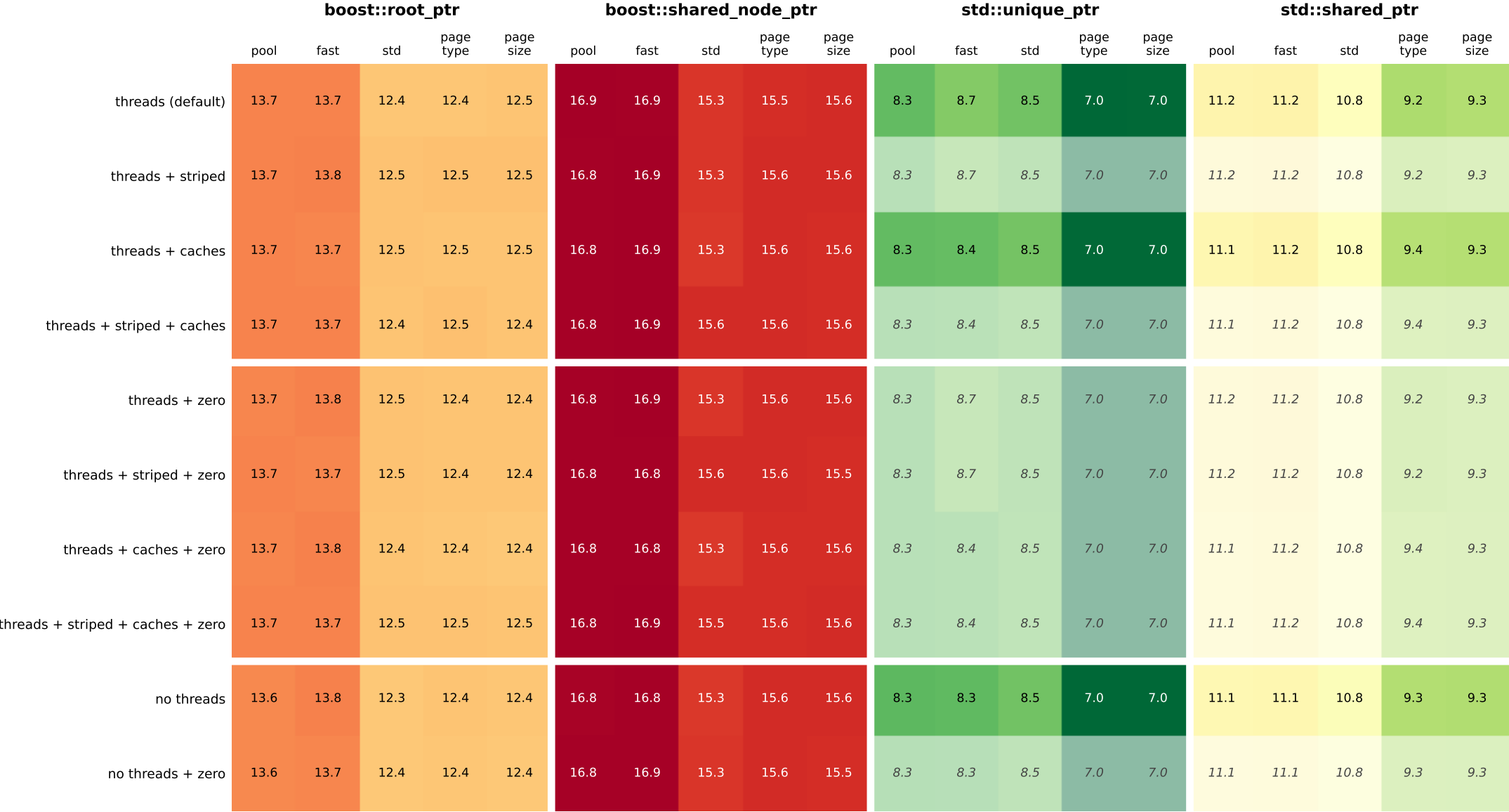
*allocator alone: one node-sized block, no pointer - churn: allocate and free one 72-byte block, 4,000,000 times; batch: allocate 100,000 blocks, then free them in order*



threads: built with thread support (the default) no threads: BOOST\_DISABLE\_THREADS striped: BOOST\_ROOT\_PTR\_STRIPED\_LOCKS caches: BOOST\_PAGE\_ALLOCATOR\_THREAD\_CACHE zero: BOOST\_ZEROIZATION  
Faded italic cells: unique\_ptr, shared\_ptr and the allocator alone have no node and no root\_ptr lock - striped\_locks and zeroization do not change their code - so they show the equivalent build's measurement.

Resident memory grown while the objects are live: bulk (lower is better)

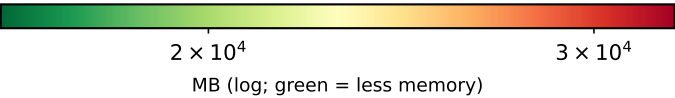
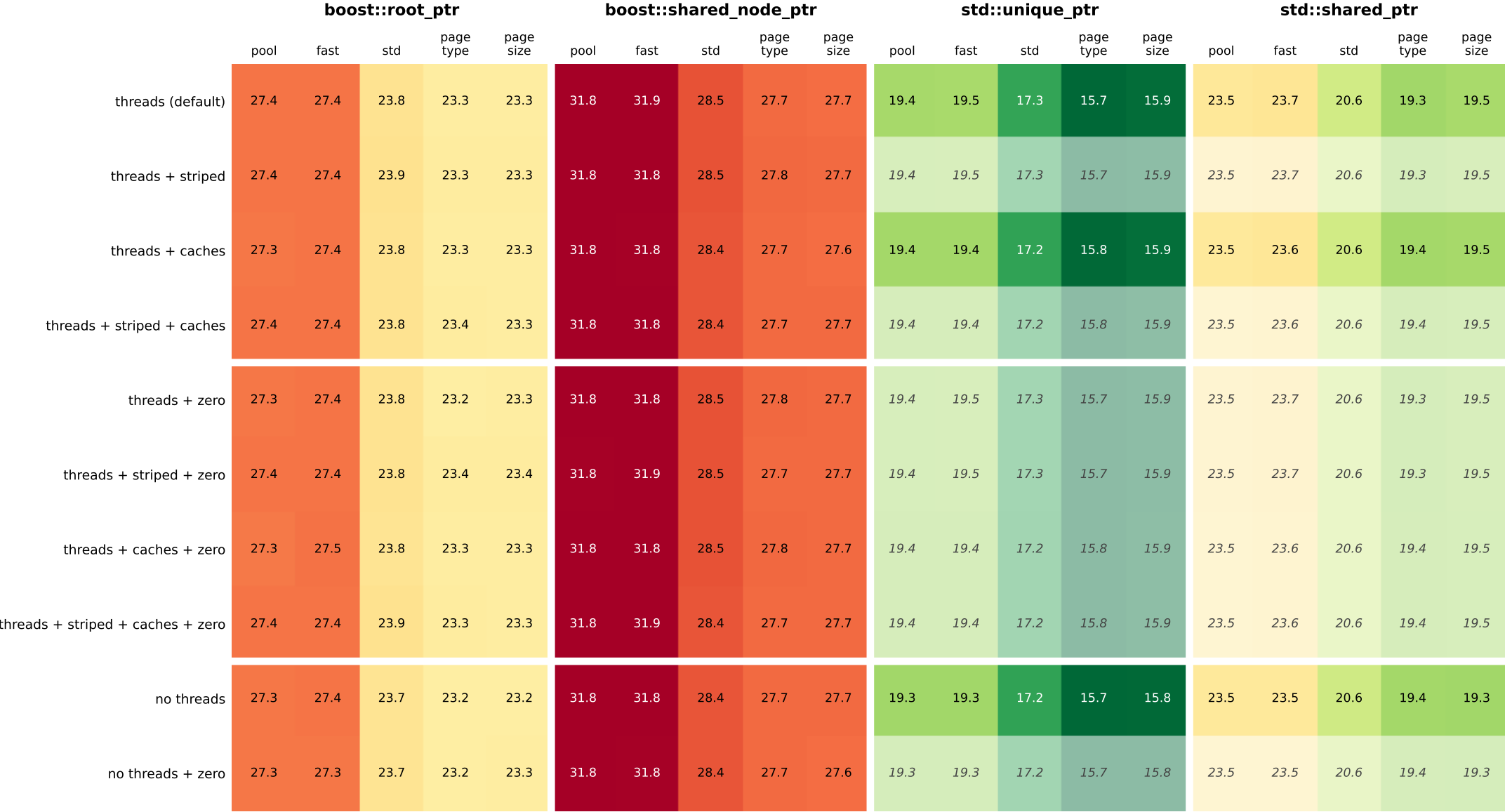
100,000 live 48-byte objects, released together - MB, median of 5 runs



threads: built with thread support (the default) no threads: BOOST\_DISABLE\_THREADS striped: BOOST\_ROOT\_PTR\_STRIPED\_LOCKS caches: BOOST\_PAGE\_ALLOCATOR\_THREAD\_CACHE zero: BOOST\_ZEROIZATION  
Faded italic cells: unique\_ptr, shared\_ptr and the allocator alone have no node and no root\_ptr lock - striped locks and zeroization do not change their code - so they show the equivalent build's measurement.

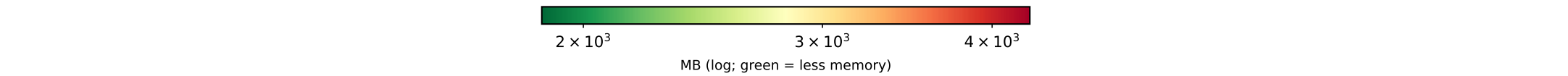
Resident memory grown while the objects are live: mixed (lower is better)

50,000 each of 16, 48 and 200 bytes, interleaved, released together - MB, median of 5 runs



threads: built with thread support (the default) no threads: BOOST\_DISABLE\_THREADS striped: BOOST\_ROOT\_PTR\_STRIPED\_LOCKS caches: BOOST\_PAGE\_ALLOCATOR\_THREAD\_CACHE zero: BOOST\_ZEROIZATION  
Faded italic cells: unique\_ptr, shared\_ptr and the allocator alone have no node and no root\_ptr lock - striped locks and zeroization do not change their code - so they show the equivalent build's measurement.

32 types of equal size, 100 live objects of each - MB, median of 5 runs

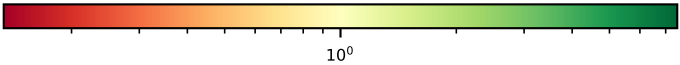


Effect of turning thread support off (BOOST\_DISABLE\_THREADS) (higher is better)

time without / time with the setting, each pair of builds differing by it alone: above 1.00x the setting is faster, below it is slower

|                               | default |       |       |           |           | with zero |       |       |           |           |
|-------------------------------|---------|-------|-------|-----------|-----------|-----------|-------|-------|-----------|-----------|
|                               | pool    | fast  | std   | page type | page size | pool      | fast  | std   | page type | page size |
| allocator alone: raw_churn    | 1.57x   | 5.28x | 1.09x | 4.18x     | 3.59x     | 1.57x     | 5.28x | 1.09x | 4.18x     | 3.59x     |
| allocator alone: raw_batch    | 0.99x   | 1.16x | 1.00x | 1.16x     | 1.15x     | 0.99x     | 1.16x | 1.00x | 1.16x     | 1.15x     |
| boost::root_ptr: churn        | 2.19x   | 5.43x | 2.29x | 6.23x     | 6.17x     | 1.90x     | 4.32x | 2.47x | 5.49x     | 5.58x     |
| boost::root_ptr: bulk         | 0.98x   | 1.70x | 1.57x | 1.76x     | 1.78x     | 0.97x     | 1.63x | 1.61x | 1.71x     | 1.69x     |
| boost::root_ptr: mixed        | 0.98x   | 1.52x | 1.37x | 1.38x     | 1.77x     | 0.98x     | 1.41x | 1.11x | 1.40x     | 1.38x     |
| boost::root_ptr: cycles       | 1.00x   | 2.35x | 2.59x | 2.84x     | 2.74x     | 0.99x     | 2.12x | 2.11x | 2.74x     | 2.83x     |
| boost::root_ptr: types        | 1.25x   | 1.36x | 1.52x | 1.17x     | 1.76x     | 1.19x     | 1.63x | 1.73x | 1.10x     | 1.68x     |
| boost::shared_node_ptr: churn | 1.56x   | 1.67x | 1.00x | 1.79x     | 2.16x     | 1.35x     | 1.37x | 1.06x | 1.74x     | 1.59x     |
| boost::shared_node_ptr: bulk  | 1.02x   | 1.21x | 1.07x | 1.26x     | 1.22x     | 1.00x     | 1.26x | 1.13x | 1.19x     | 1.24x     |
| boost::shared_node_ptr: mixed | 0.92x   | 1.19x | 0.97x | 1.07x     | 1.39x     | 1.00x     | 1.16x | 1.00x | 1.17x     | 1.13x     |
| boost::shared_node_ptr: types | 1.02x   | 1.12x | 1.08x | 0.95x     | 1.21x     | 1.05x     | 1.13x | 1.25x | 0.96x     | 1.19x     |
| std::unique_ptr: churn        | 1.69x   | 6.49x | 1.03x | 2.85x     | 3.19x     | 1.69x     | 6.49x | 1.03x | 2.85x     | 3.19x     |
| std::unique_ptr: bulk         | 0.97x   | 1.49x | 1.02x | 1.15x     | 1.16x     | 0.97x     | 1.49x | 1.02x | 1.15x     | 1.16x     |
| std::unique_ptr: mixed        | 1.01x   | 1.29x | 1.00x | 1.01x     | 1.09x     | 1.01x     | 1.29x | 1.00x | 1.01x     | 1.09x     |
| std::unique_ptr: types        | 1.05x   | 1.35x | 1.03x | 0.95x     | 1.14x     | 1.05x     | 1.35x | 1.03x | 0.95x     | 1.14x     |
| std::shared_ptr: churn        | 2.11x   | 7.50x | 0.92x | 1.92x     | 2.43x     | 2.11x     | 7.50x | 0.92x | 1.92x     | 2.43x     |
| std::shared_ptr: bulk         | 0.98x   | 1.70x | 0.97x | 1.15x     | 1.15x     | 0.98x     | 1.70x | 0.97x | 1.15x     | 1.15x     |
| std::shared_ptr: mixed        | 1.00x   | 1.44x | 1.06x | 1.08x     | 1.10x     | 1.00x     | 1.44x | 1.06x | 1.08x     | 1.10x     |
| std::shared_ptr: types        | 1.18x   | 1.63x | 0.99x | 0.90x     | 1.15x     | 1.18x     | 1.63x | 0.99x | 0.90x     | 1.15x     |

Pairs of builds, without -> with the setting: default: threads (default) -> no threads; with zero: threads + zero -> no threads + zero



time without / time with (log; green = the setting is faster)

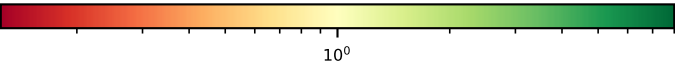
threads: built with thread support (the default) no threads: BOOST\_DISABLE\_THREADS striped: BOOST\_ROOT\_PTR\_STRIPED\_LOCKS caches: BOOST\_PAGE\_ALLOCATOR\_THREAD\_CACHE zero: BOOST\_ZEROIZATION  
'same': the pointer's code does not change with this setting. Faded italic cells: unique\_ptr, shared\_ptr and the allocator alone have no node and no root\_ptr lock - striped locks and zeroization do not change their code - so they show the equivalent build's measurement.

Effect of striped locks (BOOST\_ROOT\_PTR\_STRIPED\_LOCKS) (higher is better)

time without / time with the setting, each pair of builds differing by it alone: above 1.00x the setting is faster, below it is slower

|                                 | default |       |        |           |           | with caches |       |       |           |           | with zero |       |       |           |           | with caches + zero |       |       |           |           |
|---------------------------------|---------|-------|--------|-----------|-----------|-------------|-------|-------|-----------|-----------|-----------|-------|-------|-----------|-----------|--------------------|-------|-------|-----------|-----------|
|                                 | pool    | fast  | std    | page type | page size | pool        | fast  | std   | page type | page size | pool      | fast  | std   | page type | page size | pool               | fast  | std   | page type | page size |
| boost::root_ptr: churn          | 0.56x   | 0.45x | 0.49x  | 0.50x     | 0.48x     | 0.64x       | 0.50x | 0.50x | 0.40x     | 0.42x     | 0.55x     | 0.56x | 0.48x | 0.56x     | 0.51x     | 0.61x              | 0.51x | 0.46x | 0.42x     | 0.45x     |
| boost::root_ptr: bulk           | 1.00x   | 0.58x | 0.57x  | 0.56x     | 0.55x     | 0.98x       | 0.55x | 0.55x | 0.55x     | 0.56x     | 0.99x     | 0.57x | 0.57x | 0.56x     | 0.56x     | 1.00x              | 0.54x | 0.58x | 0.57x     | 0.57x     |
| boost::root_ptr: mixed          | 1.02x   | 0.64x | 0.64x  | 0.56x     | 0.60x     | 0.99x       | 0.64x | 0.65x | 0.57x     | 0.59x     | 0.95x     | 0.65x | 0.57x | 0.60x     | 0.59x     | 1.03x              | 0.62x | 0.65x | 0.63x     | 0.55x     |
| boost::root_ptr: cycles         | 0.99x   | 0.32x | 0.33x  | 0.31x     | 0.31x     | 1.00x       | 0.32x | 0.33x | 0.30x     | 0.29x     | 0.99x     | 0.29x | 0.34x | 0.32x     | 0.32x     | 0.99x              | 0.33x | 0.34x | 0.29x     | 0.29x     |
| boost::root_ptr: threads        | 1.43x   | 1.38x | 10.55x | 1.41x     | 1.36x     | 1.51x       | 1.33x | 8.40x | 10.02x    | 10.24x    | 1.46x     | 1.33x | 8.29x | 1.42x     | 1.35x     | 1.48x              | 1.37x | 9.46x | 9.24x     | 8.68x     |
| boost::root_ptr: types          | 0.71x   | 0.50x | 0.58x  | 0.84x     | 0.54x     | 0.73x       | 0.56x | 0.57x | 0.65x     | 0.54x     | 0.73x     | 0.58x | 0.59x | 0.81x     | 0.57x     | 0.72x              | 0.56x | 0.57x | 0.75x     | 0.53x     |
| boost::shared_node_ptr: churn   | 1.16x   | 0.97x | 0.85x  | 1.02x     | 1.22x     | 1.04x       | 0.99x | 1.13x | 1.01x     | 1.01x     | 1.06x     | 1.03x | 1.07x | 0.98x     | 1.00x     | 0.95x              | 0.99x | 0.92x | 0.96x     | 1.00x     |
| boost::shared_node_ptr: bulk    | 1.01x   | 1.01x | 0.99x  | 1.04x     | 0.99x     | 1.01x       | 1.01x | 1.05x | 1.02x     | 0.98x     | 1.00x     | 1.02x | 1.00x | 1.00x     | 0.99x     | 1.01x              | 0.99x | 1.02x | 0.99x     | 1.03x     |
| boost::shared_node_ptr: mixed   | 0.99x   | 1.01x | 0.84x  | 1.01x     | 1.12x     | 1.00x       | 1.14x | 0.98x | 0.99x     | 0.97x     | 1.02x     | 1.13x | 0.99x | 1.16x     | 1.02x     | 1.02x              | 1.12x | 1.15x | 1.06x     | 1.09x     |
| boost::shared_node_ptr: threads | 1.06x   | 1.04x | 0.82x  | 1.07x     | 1.04x     | 1.06x       | 1.06x | 0.97x | 1.17x     | 1.00x     | 1.03x     | 1.11x | 1.03x | 0.97x     | 1.01x     | 1.00x              | 1.02x | 0.83x | 1.02x     | 1.29x     |
| boost::shared_node_ptr: types   | 0.94x   | 0.99x | 0.98x  | 1.01x     | 1.00x     | 1.02x       | 0.93x | 1.00x | 0.91x     | 1.09x     | 1.02x     | 1.03x | 1.01x | 0.91x     | 1.01x     | 0.95x              | 1.02x | 0.97x | 0.97x     | 1.05x     |

Pairs of builds, without -> with the setting: default: threads (default) -> threads + striped; with caches: threads + caches -> threads + striped + caches; with zero: threads + zero -> threads + striped + zero; with caches + zero: threads + caches + zero -> threads + striped + caches + zero



time without / time with (log; green = the setting is faster)

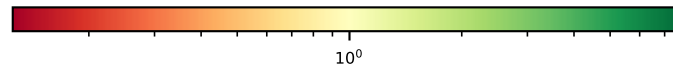
threads: built with thread support (the default) no threads: BOOST\_DISABLE\_THREADS striped: BOOST\_ROOT\_PTR\_STRIPED\_LOCKS caches: BOOST\_PAGE\_ALLOCATOR\_THREAD\_CACHE zero: BOOST\_ZEROIZATION  
'same': the pointer's code does not change with this setting. Faded italic cells: unique\_ptr, shared\_ptr and the allocator alone have no node and no root\_ptr lock - striped locks and zeroization do not change their code - so they show the equivalent build's measurement.

# Effect of per-thread caches (BOOST\_PAGE\_ALLOCATOR\_THREAD\_CACHE) (higher is better)

time without / time with the setting, each pair of builds differing by it alone: above 1.00x the setting is faster, below it is slower

|                                 | default |       |       |           |           | with striped |       |       |           |           | with zero |       |       |           |           | with striped + zero |       |       |           |           |
|---------------------------------|---------|-------|-------|-----------|-----------|--------------|-------|-------|-----------|-----------|-----------|-------|-------|-----------|-----------|---------------------|-------|-------|-----------|-----------|
|                                 | pool    | fast  | std   | page type | page size | pool         | fast  | std   | page type | page size | pool      | fast  | std   | page type | page size | pool                | fast  | std   | page type | page size |
| allocator alone: raw_churn      | 1.03x   | 1.05x | 1.08x | 3.15x     | 3.10x     | 1.03x        | 1.05x | 1.08x | 3.15x     | 3.10x     | 1.03x     | 1.05x | 1.08x | 3.15x     | 3.10x     | 1.03x               | 1.05x | 1.08x | 3.15x     | 3.10x     |
| allocator alone: raw_batch      | 0.99x   | 1.01x | 1.01x | 1.02x     | 1.05x     | 0.99x        | 1.01x | 1.01x | 1.02x     | 1.05x     | 0.99x     | 1.01x | 1.01x | 1.02x     | 1.05x     | 0.99x               | 1.01x | 1.01x | 1.02x     | 1.05x     |
| boost::root_ptr: churn          | 0.97x   | 1.00x | 0.98x | 1.31x     | 1.29x     | 1.11x        | 1.12x | 1.01x | 1.05x     | 1.13x     | 0.98x     | 1.07x | 1.02x | 1.46x     | 1.25x     | 1.07x               | 0.97x | 0.98x | 1.10x     | 1.11x     |
| boost::root_ptr: bulk           | 1.01x   | 1.01x | 1.05x | 1.00x     | 1.02x     | 0.99x        | 0.97x | 1.01x | 0.98x     | 1.03x     | 0.98x     | 1.03x | 1.00x | 0.96x     | 0.95x     | 1.00x               | 0.99x | 1.02x | 0.97x     | 0.98x     |
| boost::root_ptr: mixed          | 1.04x   | 1.05x | 0.97x | 0.95x     | 1.09x     | 1.01x        | 1.06x | 0.99x | 0.95x     | 1.07x     | 1.00x     | 1.04x | 0.86x | 0.92x     | 0.98x     | 1.08x               | 1.00x | 0.97x | 0.96x     | 0.92x     |
| boost::root_ptr: cycles         | 1.01x   | 1.01x | 1.00x | 1.05x     | 1.03x     | 1.01x        | 1.02x | 0.98x | 1.02x     | 0.97x     | 0.99x     | 1.01x | 1.02x | 1.05x     | 1.09x     | 1.00x               | 1.15x | 1.03x | 0.95x     | 1.00x     |
| boost::root_ptr: threads        | 0.96x   | 1.02x | 1.09x | 1.43x     | 1.26x     | 1.01x        | 0.98x | 0.86x | 10.19x    | 9.48x     | 1.01x     | 0.96x | 0.87x | 1.37x     | 1.47x     | 1.03x               | 0.99x | 0.99x | 8.91x     | 9.39x     |
| boost::root_ptr: types          | 0.96x   | 1.00x | 1.00x | 1.13x     | 1.02x     | 0.99x        | 1.12x | 0.98x | 0.88x     | 1.03x     | 1.05x     | 1.03x | 1.00x | 0.94x     | 1.05x     | 1.03x               | 0.99x | 0.97x | 0.87x     | 0.97x     |
| boost::shared_node_ptr: churn   | 1.21x   | 0.98x | 0.90x | 1.32x     | 1.60x     | 1.08x        | 1.00x | 1.19x | 1.31x     | 1.33x     | 1.03x     | 1.03x | 1.07x | 1.36x     | 1.29x     | 0.92x               | 0.99x | 0.93x | 1.34x     | 1.29x     |
| boost::shared_node_ptr: bulk    | 1.01x   | 1.00x | 0.96x | 0.98x     | 0.98x     | 1.01x        | 1.00x | 1.02x | 0.96x     | 0.97x     | 0.99x     | 1.04x | 1.01x | 0.95x     | 0.95x     | 1.00x               | 1.00x | 1.03x | 0.95x     | 0.98x     |
| boost::shared_node_ptr: mixed   | 0.99x   | 1.03x | 0.92x | 0.95x     | 1.14x     | 1.00x        | 1.16x | 1.07x | 0.94x     | 0.98x     | 1.00x     | 1.02x | 0.95x | 1.04x     | 0.97x     | 1.00x               | 1.02x | 1.11x | 0.95x     | 1.04x     |
| boost::shared_node_ptr: threads | 1.05x   | 1.00x | 0.84x | 18.02x    | 19.28x    | 1.05x        | 1.01x | 1.00x | 19.57x    | 18.59x    | 1.07x     | 1.10x | 1.28x | 17.46x    | 16.42x    | 1.04x               | 1.01x | 1.03x | 18.40x    | 20.85x    |
| boost::shared_node_ptr: types   | 0.97x   | 1.14x | 0.96x | 0.97x     | 1.00x     | 1.05x        | 1.07x | 0.99x | 0.89x     | 1.10x     | 1.00x     | 0.99x | 1.04x | 0.91x     | 1.00x     | 0.94x               | 0.98x | 1.00x | 0.98x     | 1.03x     |
| std::unique_ptr: churn          | 0.96x   | 0.95x | 0.88x | 2.21x     | 2.90x     | 0.96x        | 0.95x | 0.88x | 2.21x     | 2.90x     | 0.96x     | 0.95x | 0.88x | 2.21x     | 2.90x     | 0.96x               | 0.95x | 0.88x | 2.21x     | 2.90x     |
| std::unique_ptr: bulk           | 1.01x   | 1.01x | 1.00x | 1.07x     | 1.11x     | 1.01x        | 1.01x | 1.00x | 1.07x     | 1.11x     | 1.01x     | 1.01x | 1.00x | 1.07x     | 1.11x     | 1.01x               | 1.01x | 1.00x | 1.07x     | 1.11x     |
| std::unique_ptr: mixed          | 1.03x   | 1.09x | 1.03x | 0.89x     | 1.04x     | 1.03x        | 1.09x | 1.03x | 0.89x     | 1.04x     | 1.03x     | 1.09x | 1.03x | 0.89x     | 1.04x     | 1.03x               | 1.09x | 1.03x | 0.89x     | 1.04x     |
| std::unique_ptr: threads        | 1.04x   | 1.01x | 0.99x | 86.68x    | 53.18x    | 1.04x        | 1.01x | 0.99x | 86.68x    | 53.18x    | 1.04x     | 1.01x | 0.99x | 86.68x    | 53.18x    | 1.04x               | 1.01x | 0.99x | 86.68x    | 53.18x    |
| std::unique_ptr: types          | 0.96x   | 1.01x | 0.98x | 0.99x     | 1.04x     | 0.96x        | 1.01x | 0.98x | 0.99x     | 1.04x     | 0.96x     | 1.01x | 0.98x | 0.99x     | 1.04x     | 0.96x               | 1.01x | 0.98x | 0.99x     | 1.04x     |
| std::shared_ptr: churn          | 0.89x   | 0.90x | 0.81x | 1.99x     | 2.67x     | 0.89x        | 0.90x | 0.81x | 1.99x     | 2.67x     | 0.89x     | 0.90x | 0.81x | 1.99x     | 2.67x     | 0.89x               | 0.90x | 0.81x | 1.99x     | 2.67x     |
| std::shared_ptr: bulk           | 0.99x   | 1.00x | 0.97x | 1.05x     | 1.04x     | 0.99x        | 1.00x | 0.97x | 1.05x     | 1.04x     | 0.99x     | 1.00x | 0.97x | 1.05x     | 1.04x     | 0.99x               | 1.00x | 0.97x | 1.05x     | 1.04x     |
| std::shared_ptr: mixed          | 1.01x   | 1.05x | 1.17x | 0.99x     | 1.01x     | 1.01x        | 1.05x | 1.17x | 0.99x     | 1.01x     | 1.01x     | 1.05x | 1.17x | 0.99x     | 1.01x     | 1.01x               | 1.05x | 1.17x | 0.99x     | 1.01x     |
| std::shared_ptr: threads        | 0.99x   | 1.04x | 1.04x | 56.16x    | 38.28x    | 0.99x        | 1.04x | 1.04x | 56.16x    | 38.28x    | 0.99x     | 1.04x | 1.04x | 56.16x    | 38.28x    | 0.99x               | 1.04x | 1.04x | 56.16x    | 38.28x    |
| std::shared_ptr: types          | 0.98x   | 1.15x | 0.98x | 0.89x     | 1.13x     | 0.98x        | 1.15x | 0.98x | 0.89x     | 1.13x     | 0.98x     | 1.15x | 0.98x | 0.89x     | 1.13x     | 0.98x               | 1.15x | 0.98x | 0.89x     | 1.13x     |

Pairs of builds, without -> with the setting: default: threads (default) -> threads + caches; with striped: threads + striped -> threads + striped + caches; with zero: threads + zero -> threads + caches + zero; with striped + zero: threads + striped + zero -> threads + striped + caches + zero



time without / time with (log; green = the setting is faster)

threads: built with thread support (the default) no threads: BOOST\_DISABLE\_THREADS striped: BOOST\_ROOT\_PTR\_STRIPED\_LOCKS caches: BOOST\_PAGE\_ALLOCATOR\_THREAD\_CACHE zero: BOOST\_ZEROIZATION

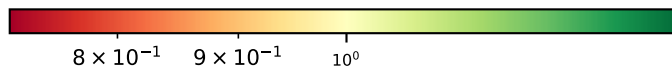
'same': the pointer's code does not change with this setting. Faded italic cells: unique\_ptr, shared\_ptr and the allocator alone have no node and no root\_ptr lock - striped locks and zeroization do not change their code - so they show the equivalent build's measurement.

# Effect of zeroization (BOOST\_ZEROIZATION) (higher is better)

time without / time with the setting, each pair of builds differing by it alone: above 1.00x the setting is faster, below it is slower

|                                 | default |       |       |           |           | with striped |       |       |           |           | with caches |       |       |           |           | with striped + caches |       |       |           |           | no threads |       |       |           |           |
|---------------------------------|---------|-------|-------|-----------|-----------|--------------|-------|-------|-----------|-----------|-------------|-------|-------|-----------|-----------|-----------------------|-------|-------|-----------|-----------|------------|-------|-------|-----------|-----------|
|                                 | pool    | fast  | std   | page type | page size | pool         | fast  | std   | page type | page size | pool        | fast  | std   | page type | page size | pool                  | fast  | std   | page type | page size | pool       | fast  | std   | page type | page size |
| boost::root_ptr: churn          | 1.03x   | 0.90x | 1.09x | 0.90x     | 0.98x     | 1.02x        | 1.12x | 1.07x | 1.01x     | 1.03x     | 1.03x       | 0.97x | 1.13x | 1.00x     | 0.95x     | 0.99x                 | 0.98x | 1.03x | 1.06x     | 1.00x     | 0.89x      | 0.72x | 1.17x | 0.79x     | 0.88x     |
| boost::root_ptr: bulk           | 1.01x   | 1.00x | 1.00x | 1.00x     | 1.03x     | 0.99x        | 0.99x | 0.99x | 1.00x     | 1.04x     | 0.98x       | 1.02x | 0.96x | 0.96x     | 0.97x     | 1.00x                 | 1.01x | 1.00x | 1.00x     | 0.99x     | 1.00x      | 0.96x | 1.03x | 0.97x     | 0.98x     |
| boost::root_ptr: mixed          | 1.01x   | 1.03x | 1.15x | 0.95x     | 1.07x     | 0.94x        | 1.05x | 1.02x | 1.01x     | 1.06x     | 0.97x       | 1.01x | 1.01x | 0.92x     | 0.96x     | 1.01x                 | 0.99x | 1.01x | 1.02x     | 0.91x     | 1.01x      | 0.95x | 0.93x | 0.96x     | 0.84x     |
| boost::root_ptr: cycles         | 1.00x   | 1.11x | 1.00x | 1.01x     | 0.94x     | 0.99x        | 1.00x | 1.00x | 1.02x     | 0.96x     | 0.99x       | 1.11x | 1.01x | 1.01x     | 0.99x     | 0.98x                 | 1.14x | 1.05x | 0.96x     | 0.99x     | 0.99x      | 1.01x | 0.81x | 0.97x     | 0.97x     |
| boost::root_ptr: threads        | 0.97x   | 1.03x | 1.12x | 0.96x     | 0.96x     | 1.00x        | 0.99x | 0.88x | 0.97x     | 0.96x     | 1.03x       | 0.96x | 0.90x | 0.92x     | 1.12x     | 1.01x                 | 0.99x | 1.01x | 0.85x     | 0.95x     | n/a        | n/a   | n/a   | n/a       | n/a       |
| boost::root_ptr: types          | 0.93x   | 0.85x | 1.00x | 1.05x     | 0.96x     | 0.97x        | 0.99x | 1.00x | 1.02x     | 1.02x     | 1.02x       | 0.88x | 1.00x | 0.87x     | 0.98x     | 1.01x                 | 0.87x | 0.99x | 1.00x     | 0.96x     | 0.89x      | 1.02x | 1.13x | 0.98x     | 0.92x     |
| boost::shared_node_ptr: churn   | 1.16x   | 0.97x | 0.94x | 1.01x     | 1.22x     | 1.06x        | 1.03x | 1.18x | 0.97x     | 1.00x     | 0.98x       | 1.02x | 1.12x | 1.04x     | 0.98x     | 0.90x                 | 1.01x | 0.92x | 0.99x     | 0.97x     | 1.00x      | 0.79x | 0.99x | 0.98x     | 0.90x     |
| boost::shared_node_ptr: bulk    | 1.00x   | 0.98x | 0.97x | 1.04x     | 1.00x     | 0.99x        | 0.99x | 0.97x | 1.00x     | 1.00x     | 0.98x       | 1.01x | 1.01x | 1.01x     | 0.97x     | 0.98x                 | 0.99x | 0.99x | 0.98x     | 1.02x     | 0.98x      | 1.02x | 1.03x | 0.98x     | 1.02x     |
| boost::shared_node_ptr: mixed   | 0.98x   | 1.03x | 0.97x | 0.90x     | 1.07x     | 1.01x        | 1.15x | 1.14x | 1.03x     | 0.97x     | 0.99x       | 1.03x | 1.00x | 0.98x     | 0.91x     | 1.00x                 | 1.02x | 1.18x | 1.05x     | 1.02x     | 1.06x      | 1.00x | 1.00x | 0.98x     | 0.86x     |
| boost::shared_node_ptr: threads | 1.00x   | 0.90x | 0.79x | 1.05x     | 1.10x     | 0.98x        | 0.96x | 0.99x | 0.94x     | 1.07x     | 1.02x       | 0.99x | 1.20x | 1.01x     | 0.93x     | 0.97x                 | 0.96x | 1.03x | 0.89x     | 1.20x     | n/a        | n/a   | n/a   | n/a       | n/a       |
| boost::shared_node_ptr: types   | 0.97x   | 0.98x | 0.96x | 0.98x     | 1.02x     | 1.05x        | 1.02x | 0.99x | 0.89x     | 1.04x     | 1.01x       | 0.85x | 1.04x | 0.92x     | 1.02x     | 0.94x                 | 0.93x | 1.00x | 0.98x     | 0.98x     | 1.01x      | 0.99x | 1.10x | 0.99x     | 1.00x     |

Pairs of builds, without -> with the setting: default: threads (default) -> threads + zero; with striped: threads + striped -> threads + striped + zero; with caches: threads + caches -> threads + caches + zero; with striped + caches: threads + striped + caches -> threads + striped + caches + zero; no threads: no threads -> no threads + zero



time without / time with (log; green = the setting is faster)

threads: built with thread support (the default) no threads: BOOST\_DISABLE\_THREADS striped: BOOST\_ROOT\_PTR\_STRIPED\_LOCKS caches: BOOST\_PAGE\_ALLOCATOR\_THREAD\_CACHE zero: BOOST\_ZEROIZATION

'same': the pointer's code does not change with this setting. Faded italic cells: unique\_ptr, shared\_ptr and the allocator alone have no node and no root\_ptr lock - striped locks and zeroization do not change their code - so they show the equivalent build's measurement.

# Fastest allocator for every build, pointer and scenario

the number: its time / the time of page\_allocator\_by\_size (the default) in the same cell - lower is better; none shown: the default is the fastest

|                                 | threads (default) | threads<br>+ striped | threads<br>+ caches | threads<br>+ striped<br>+ caches | threads<br>+ zero | threads<br>+ striped<br>+ zero | threads<br>+ caches<br>+ zero | threads<br>+ striped<br>+ caches<br>+ zero | no threads  | no threads<br>+ zero |
|---------------------------------|-------------------|----------------------|---------------------|----------------------------------|-------------------|--------------------------------|-------------------------------|--------------------------------------------|-------------|----------------------|
| allocator alone: raw_churn      | p_type 0.98       | p_type 0.98          | p_type 0.97         | p_type 0.97                      | p_type 0.98       | p_type 0.98                    | p_type 0.97                   | p_type 0.97                                | fast 0.70   | fast 0.70            |
| allocator alone: raw_batch      | p_type 1.00       | p_type 1.00          | p_size              | p_size                           | p_type 1.00       | p_type 1.00                    | p_size                        | p_size                                     | p_type 0.99 | p_type 0.99          |
| boost::root_ptr: churn          | fast 0.99         | p_type 0.98          | p_type 0.99         | p_size                           | std 0.91          | std 0.97                       | p_type 0.94                   | p_type 1.00                                | p_type 1.00 | p_size               |
| boost::root_ptr: bulk           | p_type 0.99       | p_type 0.97          | p_size              | p_size                           | p_size            | p_size                         | fast 0.99                     | p_size                                     | p_size      | p_size               |
| boost::root_ptr: mixed          | p_type 0.88       | p_type 0.94          | p_size              | p_size                           | p_type 1.00       | p_type 0.98                    | p_size                        | p_type 0.94                                | p_size      | p_type 0.99          |
| boost::root_ptr: cycles         | p_size            | p_size               | p_size              | p_type 0.98                      | fast 0.94         | p_type 0.96                    | p_type 1.00                   | fast 0.90                                  | p_type 0.99 | p_type 1.00          |
| boost::root_ptr: threads        | std 0.81          | std 0.10             | p_type 0.91         | p_type 0.93                      | std 0.69          | std 0.11                       | p_size                        | p_size                                     | n/a         | n/a                  |
| boost::root_ptr: types          | fast 0.97         | p_size               | fast 0.99           | fast 0.96                        | p_size            | p_size                         | p_size                        | p_size                                     | p_size      | p_size               |
| boost::shared_node_ptr: churn   | std 0.75          | fast 0.95            | p_size              | p_size                           | fast 0.95         | std 0.91                       | p_type 0.95                   | p_type 0.98                                | fast 0.97   | p_type 0.91          |
| boost::shared_node_ptr: bulk    | p_size            | p_type 0.99          | p_size              | p_type 0.99                      | p_type 1.00       | p_type 1.00                    | p_type 0.99                   | p_size                                     | p_type 1.00 | p_size               |
| boost::shared_node_ptr: mixed   | p_type 0.88       | p_type 0.98          | p_size              | fast 0.99                        | p_size            | p_type 0.92                    | p_type 0.98                   | fast 1.00                                  | p_size      | p_size               |
| boost::shared_node_ptr: threads | std 0.05          | std 0.07             | p_size              | p_type 0.89                      | std 0.07          | std 0.07                       | std 0.95                      | p_size                                     | n/a         | n/a                  |
| boost::shared_node_ptr: types   | p_size            | p_size               | fast 0.88           | p_size                           | p_size            | p_size                         | p_size                        | p_size                                     | p_size      | p_size               |
| std::unique_ptr: churn          | std 0.88          | std 0.88             | p_size              | p_size                           | std 0.88          | std 0.88                       | p_size                        | p_size                                     | fast 0.69   | fast 0.69            |
| std::unique_ptr: bulk           | p_type 1.00       | p_type 1.00          | p_size              | p_size                           | p_type 1.00       | p_type 1.00                    | p_size                        | p_size                                     | p_size      | p_size               |
| std::unique_ptr: mixed          | p_type 0.87       | p_type 0.87          | p_size              | p_size                           | p_type 0.87       | p_type 0.87                    | p_size                        | p_size                                     | p_type 0.93 | p_type 0.93          |
| std::unique_ptr: threads        | std 0.03          | std 0.03             | p_type 0.57         | p_type 0.57                      | std 0.03          | std 0.03                       | p_type 0.57                   | p_type 0.57                                | n/a         | n/a                  |
| std::unique_ptr: types          | p_size            | p_size               | p_size              | p_size                           | p_size            | p_size                         | p_size                        | p_size                                     | fast 0.96   | fast 0.96            |
| std::shared_ptr: churn          | std 0.78          | std 0.78             | p_size              | p_size                           | std 0.78          | std 0.78                       | p_size                        | p_size                                     | fast 0.62   | fast 0.62            |
| std::shared_ptr: bulk           | p_size            | p_size               | p_type 1.00         | p_type 1.00                      | p_size            | p_size                         | p_type 1.00                   | p_type 1.00                                | p_size      | p_size               |
| std::shared_ptr: mixed          | p_type 0.98       | p_type 0.98          | p_type 1.00         | p_type 1.00                      | p_type 0.98       | p_type 0.98                    | p_type 1.00                   | p_type 1.00                                | p_size      | p_size               |
| std::shared_ptr: threads        | std 0.04          | std 0.04             | p_type 0.62         | p_type 0.62                      | std 0.04          | std 0.04                       | p_type 0.62                   | p_type 0.62                                | n/a         | n/a                  |
| std::shared_ptr: types          | p_size            | p_size               | p_size              | p_size                           | p_size            | p_size                         | p_size                        | p_size                                     | fast 0.92   | fast 0.92            |

 boost::pool\_allocator    boost::fast\_pool\_allocator    std::allocator    boost::page\_allocator\_by\_type    boost::page\_allocator\_by\_size (default)

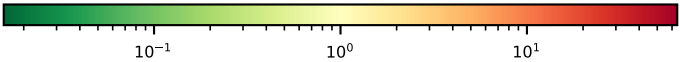
threads: built with thread support (the default)   no threads: BOOST\_DISABLE\_THREADS   striped: BOOST\_ROOT\_PTR\_STRIPED\_LOCKS   caches: BOOST\_PAGE\_ALLOCATOR\_THREAD\_CACHE   zero: BOOST\_ZEROIZATION  
p\_type: page\_allocator\_by\_type. Faded italic cells: unique\_ptr, shared\_ptr and the allocator alone have no node and no root\_ptr lock - striped locks and zeroization do not change their code - so they show the equivalent build's measurement.

The other pointers against boost::root\_ptr (lower is better)

time of the pointer / time of boost::root\_ptr in the same build, allocator and scenario: below 1.00x the pointer is faster than root\_ptr

| with boost::page_allocator_by_size (default) |  | churn       |        |        | bulk        |        |        | mixed       |        |        | threads     |        |        | types       |        |        |
|----------------------------------------------|--|-------------|--------|--------|-------------|--------|--------|-------------|--------|--------|-------------|--------|--------|-------------|--------|--------|
|                                              |  | shared node | unique | shared | shared node | unique | shared | shared node | unique | shared | shared node | unique | shared | shared node | unique | shared |
| threads (default)                            |  | 1.07x       | 0.32x  | 0.39x  | 1.06x       | 0.36x  | 0.48x  | 1.14x       | 0.47x  | 0.59x  | 0.61x       | 0.47x  | 0.43x  | 1.15x       | 0.45x  | 0.59x  |
| threads + striped                            |  | 0.42x       | 0.15x  | 0.19x  | 0.60x       | 0.20x  | 0.26x  | 0.61x       | 0.28x  | 0.35x  | 0.80x       | 0.64x  | 0.58x  | 0.63x       | 0.24x  | 0.32x  |
| threads + caches                             |  | 0.86x       | 0.14x  | 0.19x  | 1.11x       | 0.33x  | 0.47x  | 1.09x       | 0.49x  | 0.64x  | 0.04x       | 0.01x  | 0.01x  | 1.17x       | 0.44x  | 0.53x  |
| threads + striped + caches                   |  | 0.36x       | 0.06x  | 0.08x  | 0.64x       | 0.18x  | 0.26x  | 0.66x       | 0.29x  | 0.37x  | 0.41x       | 0.11x  | 0.14x  | 0.59x       | 0.24x  | 0.29x  |
| threads + zero                               |  | 0.85x       | 0.31x  | 0.38x  | 1.10x       | 0.37x  | 0.49x  | 1.14x       | 0.50x  | 0.63x  | 0.54x       | 0.45x  | 0.41x  | 1.08x       | 0.43x  | 0.57x  |
| threads + striped + zero                     |  | 0.44x       | 0.16x  | 0.19x  | 0.61x       | 0.21x  | 0.27x  | 0.66x       | 0.30x  | 0.37x  | 0.72x       | 0.61x  | 0.56x  | 0.61x       | 0.25x  | 0.32x  |
| threads + caches + zero                      |  | 0.83x       | 0.14x  | 0.18x  | 1.11x       | 0.32x  | 0.45x  | 1.15x       | 0.48x  | 0.62x  | 0.05x       | 0.01x  | 0.02x  | 1.14x       | 0.44x  | 0.53x  |
| threads + striped + caches + zero            |  | 0.37x       | 0.06x  | 0.08x  | 0.61x       | 0.18x  | 0.26x  | 0.59x       | 0.26x  | 0.34x  | 0.32x       | 0.11x  | 0.14x  | 0.58x       | 0.23x  | 0.28x  |
| no threads                                   |  | 3.05x       | 0.62x  | 0.99x  | 1.56x       | 0.55x  | 0.74x  | 1.45x       | 0.77x  | 0.95x  | n/a         | n/a    | n/a    | 1.66x       | 0.69x  | 0.90x  |
| no threads + zero                            |  | 3.00x       | 0.55x  | 0.88x  | 1.50x       | 0.54x  | 0.72x  | 1.40x       | 0.64x  | 0.79x  | n/a         | n/a    | n/a    | 1.53x       | 0.64x  | 0.83x  |

| with std::allocator               |  | churn       |        |        | bulk        |        |        | mixed       |        |        | threads     |        |        | types       |        |        |
|-----------------------------------|--|-------------|--------|--------|-------------|--------|--------|-------------|--------|--------|-------------|--------|--------|-------------|--------|--------|
|                                   |  | shared node | unique | shared | shared node | unique | shared | shared node | unique | shared | shared node | unique | shared | shared node | unique | shared |
| threads (default)                 |  | 0.79x       | 0.28x  | 0.30x  | 1.04x       | 0.46x  | 0.53x  | 1.08x       | 0.53x  | 0.69x  | 0.04x       | 0.02x  | 0.02x  | 1.06x       | 0.47x  | 0.55x  |
| threads + striped                 |  | 0.45x       | 0.14x  | 0.15x  | 0.60x       | 0.26x  | 0.31x  | 0.83x       | 0.34x  | 0.44x  | 0.52x       | 0.17x  | 0.20x  | 0.63x       | 0.27x  | 0.32x  |
| threads + caches                  |  | 0.86x       | 0.31x  | 0.36x  | 1.13x       | 0.48x  | 0.57x  | 1.15x       | 0.50x  | 0.57x  | 0.05x       | 0.02x  | 0.02x  | 1.11x       | 0.48x  | 0.56x  |
| threads + striped + caches        |  | 0.39x       | 0.16x  | 0.18x  | 0.60x       | 0.26x  | 0.32x  | 0.76x       | 0.32x  | 0.37x  | 0.45x       | 0.15x  | 0.17x  | 0.63x       | 0.28x  | 0.32x  |
| threads + zero                    |  | 0.91x       | 0.30x  | 0.33x  | 1.07x       | 0.46x  | 0.54x  | 1.29x       | 0.61x  | 0.79x  | 0.06x       | 0.02x  | 0.02x  | 1.10x       | 0.47x  | 0.55x  |
| threads + striped + zero          |  | 0.41x       | 0.14x  | 0.16x  | 0.61x       | 0.26x  | 0.30x  | 0.74x       | 0.35x  | 0.45x  | 0.46x       | 0.15x  | 0.18x  | 0.64x       | 0.28x  | 0.32x  |
| threads + caches + zero           |  | 0.87x       | 0.35x  | 0.41x  | 1.07x       | 0.46x  | 0.55x  | 1.16x       | 0.51x  | 0.58x  | 0.04x       | 0.02x  | 0.02x  | 1.06x       | 0.48x  | 0.56x  |
| threads + striped + caches + zero |  | 0.43x       | 0.16x  | 0.19x  | 0.60x       | 0.26x  | 0.32x  | 0.65x       | 0.33x  | 0.37x  | 0.44x       | 0.15x  | 0.17x  | 0.62x       | 0.27x  | 0.32x  |
| no threads                        |  | 1.80x       | 0.62x  | 0.74x  | 1.53x       | 0.71x  | 0.87x  | 1.54x       | 0.72x  | 0.89x  | n/a         | n/a    | n/a    | 1.49x       | 0.70x  | 0.85x  |
| no threads + zero                 |  | 2.13x       | 0.72x  | 0.87x  | 1.53x       | 0.72x  | 0.89x  | 1.43x       | 0.67x  | 0.83x  | n/a         | n/a    | n/a    | 1.53x       | 0.79x  | 0.96x  |



threads: built with thread support (the default) no threads: BOOST\_DISABLE\_THREADS striped: BOOST\_ROOT\_PTR\_STRIPED\_LOCKS caches: BOOST\_PAGE\_ALLOCATOR\_THREAD\_CACHE zero: BOOST\_ZEROIZATION  
Faded italic cells: unique\_ptr, shared\_ptr and the allocator alone have no node and no root\_ptr lock - striped locks and zeroization do not change their code - so they show the equivalent build's measurement. Cycles: only root\_ptr reclaims them.

# Bulk release scaling, threads (default) build (lower is better)

ns per object, one run per size: boost::pool\_allocator grows with the number of objects, the others stay flat

